Stiffness as a Discrimination Criterion to Select Harvestable Tea Shoots

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Abstract. Selection is an essential part of the tea harvesting process, which effectively determines the quality of made tea. Only mature shoots on a tea bush are selected to be harvested while leaving the immature ones intact. Colour, size, texture, height and stiffness of tea shoots are commonly used properties to distinguish harvestable tea shoots. In this research, stiffness and height of tea shoots were identified as two parameters having potential for mechanical-form of selection. To this end, both height and stiffness of tea shoots were measured randomly and collected data was used to statistically verify the appropriateness of the two parameters for discrimination. As an initial step, only stiffness-based selection was considered. In order to make the selection, the deflection of a plastic strip with a predetermined stiffness when it was pushed against the tea shoots was taken into account. The deflection of the strip as it encountered an immature shoot was found to be less than that of a matured one. A working model which uses this physical phenomenon was constructed to test the effectiveness of the stiffness-based selection concept. The first prototype was designed, fabricated and tested on field. Several key parameters of the 'tea shoot selective mechanism' were determined and researched upon. Consequently with the understanding on the performance of the first prototype, several design changes were incorporated into the next generations of prototypes in an iterative manner. A performance indicator known as 'selectivity index' was formulated to compare the ability of those different selection mechanisms. Finally, drawbacks of the tested prototypes were identified for future improvement.

Keywords: selective tea plucking, stiffness-based selection, selectivity index

Stiffness as a discrimination criterion to select harvestable tea shoots

1 Introduction

Sri Lanka had a share of 17.3% of world tea exports in 2011 [1]. Total tea exports amounted to Rs. 165 Billion, bringing the second largest income for the country in 2011 [1]. Furthermore, it was the major agricultural export of the country, which earned the largest income from exports in the latter year [1]. According to [1] Central and Southern are the provinces that produce the most of the made tea in Sri Lanka. To illustrate this, the two provinces accounted for 33% and 28% of the total tea production in 2011 [1].

Tea production involves several stages, i.e. tea plucking, weighing, withering, rolling, aeration, drying, grading and packaging [2]. In Sri Lankan tea industry the workers pluck tea leaves and weigh them on weighing scales [2]. Withering is the subsequent process of removing some amount of moisture from tea leaves and machines called troughs are used for the process [2]. Afterwards, rolling is performed with rolling scales [2]. In this procedure, leaf cells are ruptured to release enzymes and to bring them into contact with air so that aeration can commence [2]. During the aeration process, chemical reactions take place [2]. For this, fermentation machines are used [3]. Likewise, drying, grading and packaging phases are also wholly automated [2] [3]. However, automation has not been incorporated into the tea plucking phase in Sri Lanka.

In order to produce high quality tea, leaves which are harvested must be at the correct maturity level. Tea shoot is a stem where couple of tea leaves and a bud are grown on the top-most point of it [4]. Tender shoots having 2-3 immature leaves and/or dormant buds should be plucked to produce high quality tea [5]. In a typical tea bush, there are several generations of tea shoots on its plucking table, where a generation is considered as a family of tea shoots that are at the same level of growth [4]. There are 5-6 distinguished shoot generations found to be in a typical tea bush in Sri Lanka [6]. In a single harvesting round, one or two older generations are plucked, leaving the younger ones intact [5]. This action is defined as '*Selection'*. The younger generations are called 'arimbu'. Harvesting 'arimbu' results in reduction of productivity of the tea bush [6]. Meeting the above requirements requires cumbersome labour. A skilled tea-harvester can pluck up to 20 kg of leaves daily [2]. If workers miss selecting the harvestable shoots, they will become too matured by the time of the next plucking round. As a result, if they are being plucked in the successive plucking round, the quality of tea will be reduced. Furthermore, such mature shoots will reduce the yield of the tea bush.

Average daily wage of a male tea harvester in 2012 was Rs. 548 and that of a female was Rs. 435 [1]. This relative high cost of labour has pushed the industry towards mechanisation of tea harvesting, and several tea harvesting machines has been devel-

oped. However, a common drawback of these machines is the lack of selectivity, which results in loss of crop [4].
This research aimed to find and validate a method to select harvestable shoots from the set of different generations of shoots in local tea bushes. A selection method should ideally pluck all the harvestable shoots on the bush while leaving all the immature arimbus unharmed.

2 Research Methodology and Materials

First objective was to determine criteria in order to perform selection. Several criteria for selection were identified or formulated using literature and the wealth of experience of tea pluckers. Afterwards, these criteria were tested to see whether they can be used to achieve selectivity.
According to experience of tea pluckers; colour, size, texture, height and stiffness of tea shoots are used to discriminate between arimbus and harvestable shoots. Electronic sensor-based selection methods are required for the first three selection criteria. The ambience of tea fields is in general wet and unpredictable. Therefore, electronic sensors can become an unreliable option especially in the field. Latter two criteria which can be exercised by mechanical means provide a more pragmatic approach. Therefore, these two were selected for further investigations.
Tea pluckers pluck tea shoots from an imaginary surface on tea bush, which is referred as the 'plucking table'. Heights of harvestable tea shoots and arimbus were measured randomly from the plucking table in a tea estate that cultivates TRI 2026 clone tea. The heights were measured from the plucking table. A statistical analysis was then performed on the data using Minitab (a statistical software for analysing data) to check whether heights of arimbus and harvestable shoots are significantly different. Afterwards, the stiffness values of tea shoots were measured and a similar statistical analysis was performed to check for differences between harvestable and arimbu tea shoots in terms of stiffness. A custom-made test apparatus was developed for the measurement of stiffness as depicted in Fig.1.

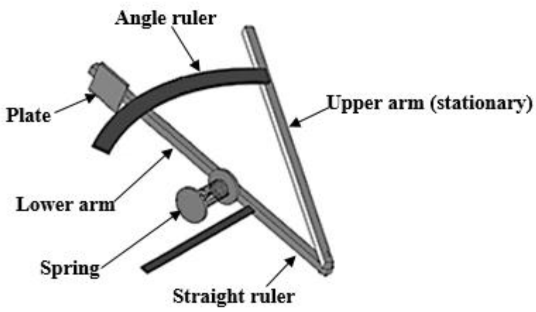

Fig. 1. Apparatus constructed to measure stiffness of tea shoots.

Initially, the plate was held adjacent to a tea shoot at a predetermined height from the plucking table. Secondly, the plate was pushed against the tea shoot incrementally to minimise the inertial effects of the loadings. As the tea shoot got deflected, the amount of deflection was measured with a protractor named angle-ruler. The force required was calculated by observing the extension of the spring attached to the lower arm using the straight-ruler.

With the data obtained, two multiple regression models were built for arimbus and harvestable tea shoots. The independent variables used in this analysis were height from plucking table and the amount of deflection. The dependent variable was the force required for the deflection which has been defined as 'stiffness'. Such regression model is helpful in cases where stiffness values for different levels of deflections at various heights from the plucking table are required for future designs. Afterwards, stiffness values were calculated for range of heights: 0 mm to 100 mm in steps of 5 mm. At each height, a statistical analysis was performed to check whether stiffness of arimbus and harvestable shoots were significantly different.

Next step was to design and develop effective selection mechanisms that can perform selection comprehensively and subsequently to test them on field to verify their practical applicability. It was considered that the deflection of a flexible strip can be used to select tea shoots. The characteristic that was used to discriminate arimbus from harvestable shoots was the amount of deflection that the strips undergo upon contact. The stiffness values of plastic strips of varying lengths were measured using the custom-made equipment shown in Figure 2. The width of each strip was 10 mm with a thickness of 1 mm.

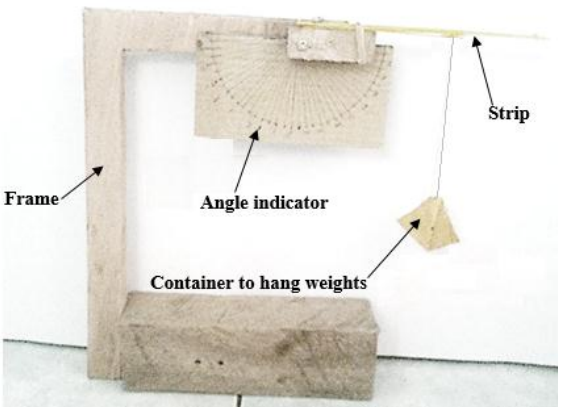

Fig. 2. Apparatus to measure stiffness of plastic strips

In order to test and study the effectiveness of the proposed concept, a working model was designed and fabricated to be used in the field, which will be referred to as the first prototype hereafter. The configuration of the apparatus is shown in Fig.3.

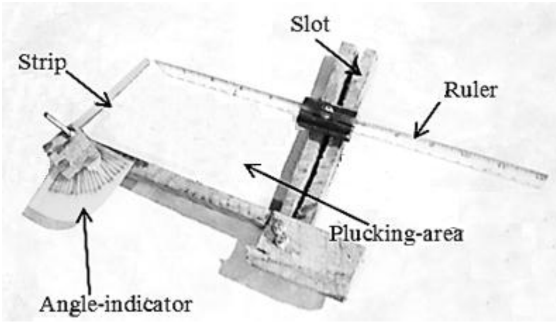

Fig. 3. The first prototype

The prototype was traversed over the tea bush approximately 10 mm above the plucking table with respect to the plastic strip which is responsible for performing the selection. The height for selection of the tea shoots was determined to minimize the interference of the mature tea shoots with plastic strip observed randomly over the plucking table. Through the first experiment, it was also revealed that with the increase of height from plucking table, the difference of stiffness of arimbus and harvestable shoots diminish rapidly. Therefore, a suitable as well as detectable height was determined, which served as the best compromise for the identified test conditions. The direction of movement was parallel to the ruler and when an arimbu comes into contact with the strip, it was supposed to slide along the strip till it goes beyond the ruler. When a harvestable shoot turns up, it was supposed to bend the strip and go inside the plucking-area.

In the first prototype, angle of inclination of the strip to the direction of traversal and length of the plastic strip can be adjusted as shown in Fig. 3. Several experiments were then performed in a tea field while changing the above mentioned parameters to evaluate their sensitivity.

When performing selective tea harvesting with a certain mechanism, a predetermined region on a tea bush was selected and the number of arimbus and harvestables were counted. Then the selection mechanism was traversed over the identified area and in the process correctly selected harvestable shoots together with the erroneously selected arimbus that gathered into the plucking area of the mechanism were counted. A performance indicator known as the 'selectivity index' (s) was formulated as indicated in equation 1 to quantify the effectiveness of the selection mechanism. This index will also provide a platform to identify the influence of adjustable variables in the prototype. Moreover, it will serve as a reference for comparing and contrasting several selective mechanisms that will be derived iteratively with subsequent improvements.

$$s = 10^{1+\frac{h}{H}-\frac{a}{A}} \tag{1}$$

Where,

H - Number of harvestables in the area before selection

h – Number of harvestables that were selected to be harvested

A – Number of arimbus before selection

a – Number of arimbus that are selected (erroneously) to be harvested

As the selections mechanism's effectiveness improve, the selectivity index (s), increase concurrently. This index ranges from 1 where all arimbus and no harvestables are selected to 100 where no arimbus and all the harvestables are selected. For the statistical analyses, arimbus and harvestable shoots were taken as two populations. In order to implement a comprehensive selection mechanism, arimbus and harvestable shoots need to have distinctive heights and stiffness values. To verify this, students' T tests for difference between means were carried out for both stiffness and height. T-tests for height and stiffness showed that the two populations were significantly distinct with a p-value of 0.00.

3 Design, Results and Evaluation

As described in the methodology, stiffness alone was chosen as the selection criterion for selective plucking. The first prototype was intended to be used as a stiffness-based selection mechanism. The elastic modulus of the strip being used was found to be 1.6 GPa in the experiment shown in Fig.2. This value was comparable with that of ABS plastics where the modulus ranges from 1.1 to 2.9 GPa [7]. By regression models built for stiffness of tea shoots, the mean value for deflecting a harvestable shoot by 10 mm at a height of 10 mm was 5.65 g. Similarly, for arimbu, the value was 0.56 g. Therefore selection was decided to be performed at the mid-point of these two at 3.1 g. However, if the plastic strip poses too much restriction, tea shoots could well be subjected to severe deflection and result in crawling under the strip into the plucking area by-passing the proper selection process. In order to resolve this situation, strip was oriented at an angle to the direction of traversal. By the experiment carried out (as shown in Fig. 2), the force required to deflect a plastic strip of length 60mm having an inclined angle of 135^0 to the direction of traversal with 10 mm of deflection was found to be 3.1g.

The optimum value for the length of the plastic strip and angle of inclination to the direction of traversal had to be determined with the view of maximising the effectiveness of the selection mechanism. To perform this, an analytical approach was considered first with respect to the length. The deflection at the free end of the strip can be expressed by equation 2.

$$Vp = \frac{F}{3EI}[\cos(\alpha_1) + \mu\sin(\alpha_1)]\, l^3 \qquad (2)$$

Where,

V_p – Deflection of the free end of the strip
α_1 – Angle of the tangent at the free end of the strip to its initial position
l – Length of the strip
E – Elastic modulus
I – Second moment of area

Here F is the mean force required to deflect harvestable shoots and arimbus with a deflection of x magnitude. This can be represented using the expression in equation 3.

$$F = (2.008 + 0.11x) * 9.81 * 10^{-3} \qquad (3)$$

The selection process naturally becomes more comprehensive when V_p increases. By complementing the data obtained here and those obtained earlier experiments, it was concluded mathematically that with a plastic strip of length 70 mm, the task of selection can be achieved considering angle of attack of a harvestable shoot is perpendicular to the strip. It was also observed that, as the strip length increased, the number of shoots getting engaged with the selection mechanism tends to vary significantly. This is mainly due to the nonhomogeneous distribution of shoots at the plucking table. Therefore, the practical limit on length of strip had to be considered. As previously mentioned with Fig. 3, the inclination of the strip to the direction of traversal could be changed. Through experimentation it was observed that shoots will experience lesser deflection as the angle of inclination increased. This phenomenon was especially useful to prevent arimbus from crawling under the strip into the cutting area. In addition, as the inclination angle is increased the effective width of the selection mechanism was reduced to prevent multiple shoots coming into engagement with the strip. A series of experiments were then arranged to determine the sensitivity of the two adjustable variables discussed above. While changing the variable values in a systematic manner, selectivity index was calculated for discrete number of test steps, wherein the maximum selectivity index of 20 was obtained for a length of 70 mm and at an inclined angle of 160^0. Interestingly, these experimental results were found to be in agreement with the mathematical analysis.
Thereafter, with the understanding of the performance of the first prototype, several design changes were incorporated into subsequent prototypes in an iterative manner. Thus, various selection mechanisms were developed and tested, and compared using the selectivity index. The key design issues with respect to this original development can be summarised as follows. By design, arimbus were supposed to slide along the strips and be prevented from entering the cutting-area. However, when arimbus are too short, they tend to crawl under the strips and slip into the cutting area erroneously. In addition, when the strip encounters a bunch of tea shoots, the apparent stiffness of those shoots increases and they arrive at the cutting area.

4 Future Work

Issues experienced on existing designs needs to be resolved with suitable design modifications. The work is still in the experimental phase and need to be converted to a form that can be used in conjunction with a commonly used harvester for proper field testing. In these designs, only stiffness was considered as the selection criterion. However, height of tea shoots is another plausible criterion as seen in previous statistical analyses. Therefore, designs which also incorporate height as a selection criterion have to be studied. Furthermore, selectivity index was formulated to compare selectivity of various selection methods. However, this index requires experimental validation.

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